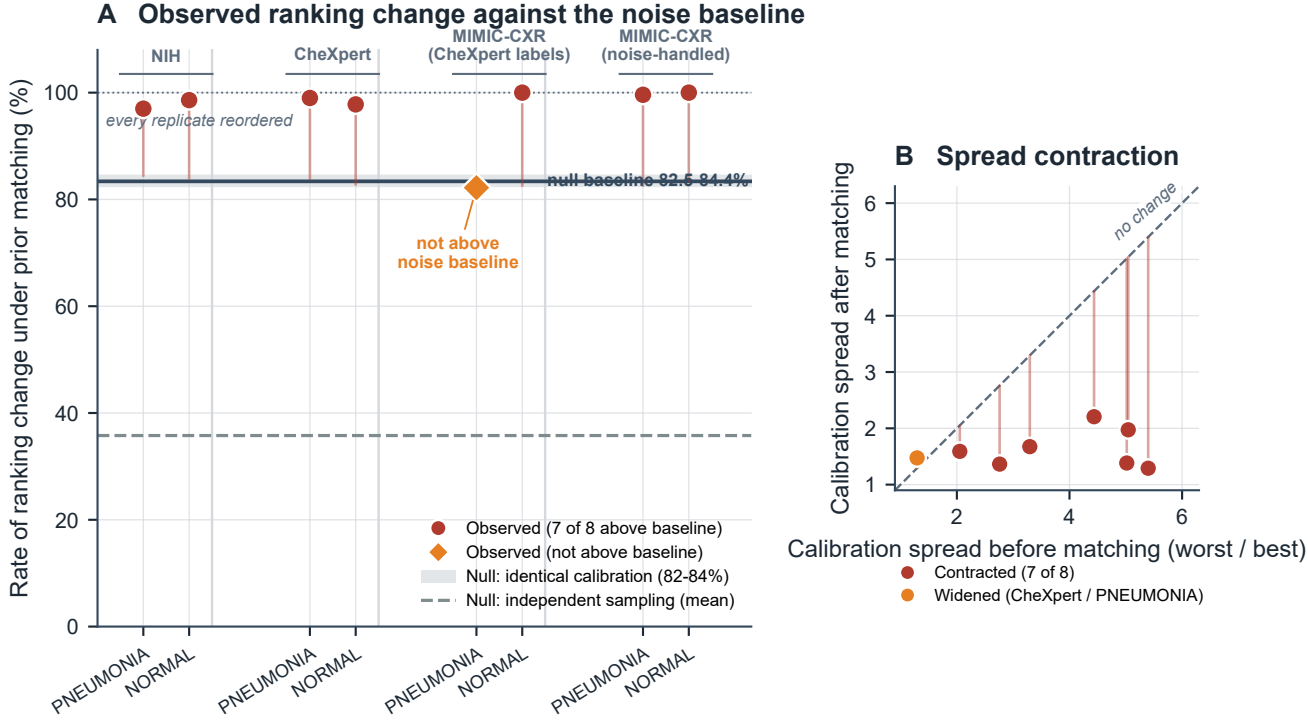


Panel A: rate at which prior matching reordered the three datasets by calibration, against null distributions that resample under the same protocol. Under a null of identical calibration across datasets (shaded band), prior matching alone reordered them in 82-84% of replicates; the observed rate exceeded this baseline in 7 of 8 model-label combinations, by a median of 15.1 percentage points (dataset-level sign-flip test, one-sided p = 0.031; 2^8 = 256 sign flips, minimum attainable p = 0.0039). In the single exception (MIMIC-CXR with CheXpert labels, PNEUMONIA) the observed rate fell below the baseline. Panel B: calibration spread before (horizontal axis) and after (vertical axis) prevalence matching; points below the dashed no-change line indicate contraction. The largest contraction is MIMIC-CXR (noise-handled) on NORMAL (23.9% of the original spread retained).